

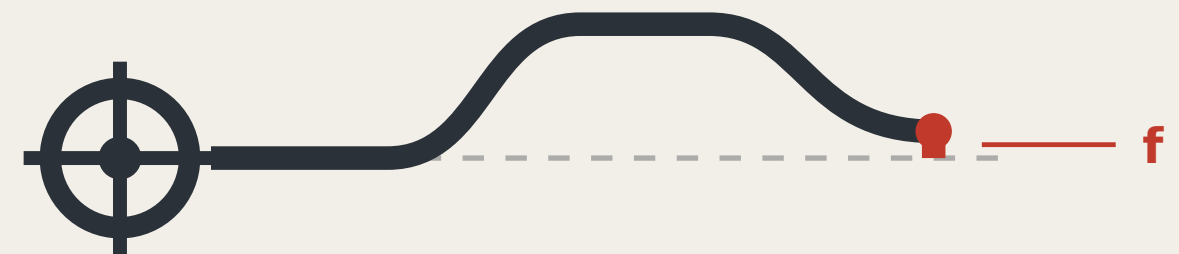
THE LEVELING LINE

A3 BOOKLET

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.
Depart, carry the height station by station, return — that gap is f .
Within tolerance, every station on the line is accepted;
over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:
what a city's trust in its AI rests on: not performance, but return.



Closure error f

Two readings of the same point, and the trust metric for this corridor.

CONTENTS

FIG.00	The Leveling Line: a city that publishes its own error	FIG.17	The step-free link from the origin community to the heritage park
FIG.01	Overall concept and site cross-check	FIG.18	Maintenance: putting the wear on the parts that can be swapped
FIG.02	Evidence chain and submission package: a leveling circuit not yet closed	FIG.19	Monuments beside heritage fabric: the no-drilling construction and its ...
FIG.03	Three scope levels and network orders	FIG.20	Reading after dark, without lighting the benchmark
FIG.04	Three key areas and benchmark layout	FIG.21	How far the nearest benchmark actually is: this proposal's own rule, ap...
FIG.05	Slow mobility, blue-green and closing routes	FIG.22	Winter: water, ice, and two of this package's rules in collision
FIG.06	Recomputed metrics and the field census	FIG.23	Finding the nearest benchmark: which way, and how far
FIG.07	Identity: mark, construction and applications	FIG.24	The device envelope: what exactly goes in those 18 m ²
FIG.08	Innovation ecosystem and element mechanisms	FIG.25	BM-0, the origin: the place both routes have to return to
FIG.09	Landmarks, kit of parts, signage syntax and operating cycle	FIG.26	BM-2x, the second order: a scenario has to be stoppable, and that take...
FIG.10	The benchmark on the street, and how the kerb is shared	FIG.27	One reading: where a person stands, and whether they block the way
FIG.11	Regional coordination interfaces: extending the levelling network	FIG.28	Third order: the most numerous tier, and more than one reader
FIG.12	Four-quadrant pedestrian connection and device queue reservoir at D...	FIG.29	The annual first-order closure: how long a route actually takes
FIG.13	The three key areas in section, at one scale	FIG.30	The year: where forty-seven readings fall
FIG.14	The through-block public route to the Qing river	FIG.31	The other year: the hours this network asks of people it does not pay
FIG.15	Phasing: advanced by closure results, not by dates	FIG.32	How many people: the cheapest roster is the one that empties the instrum...
FIG.16	The benchmark and its reading plate, as a construction detail		

a type drawing; it carries no orientation

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

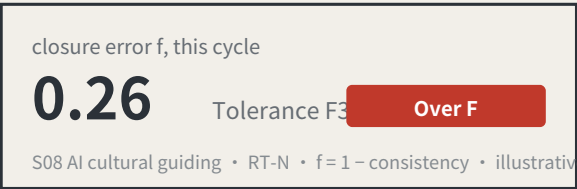
BM-0 • AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

the L3 return-to-datum point

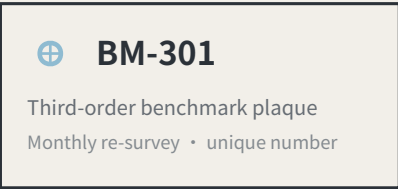
BM-2 • Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

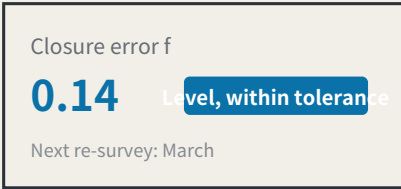
Kit of parts for public space: five standard components

KIT OF PARTS • common specifications and open drawings, so any new node joins in one language



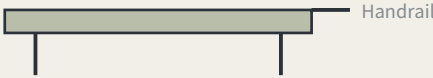
1 Stone and plaque

The number is the position, the order and the cycle Visible on site; no internet needed



2 Reading board

Visible on site; no internet needed



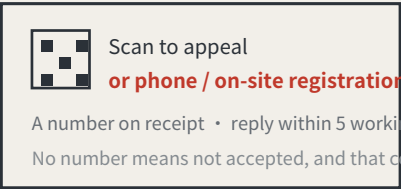
3 Dwellable seating

With armrests: older people rely on them to stand



4 Step-free guidance

Read by the users themselves



5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	Origin benchmark	Annual origin and closure computation
BM-1 / BM-2	First-order benchmark	Annual re-survey
BM-2x	2nd-order point	Quarterly re-survey
BM-3xx	3rd-order point	Monthly re-survey
RT-N / RT-S	Closing route	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	Community day	third order BM-301/BM-302/BM-303 • P5 • P7
Quarterly	Scenario open day	second order BM-21/BM-22 P2 • P3
Semi-annual	Route re-survey	first order, routes RT-N / RT-S P1 • P2
Annual	Zeroing ceremony	the L3 return-to-datum point all four review parties present

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	BM-21	→	Deliberately blank — see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.

Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Id	Case	Trust mechanism	Re-measurable	Transferable point
C1	Algorithm registers	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	Risk-tiered AI legislation	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	Standardised testing frameworks	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	Algorithmic impact assessment practice	Mandatory ex-ante and ex-post review for high-risk uses	led-high	Supports the strictest tolerance for health scenarios
C5	Civic data-stewardship practice	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	Open-source reproducibility norms	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

Method, two measurements, and the boundaries of application

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leq F$ is always the passing test.

Act One: measure this open call first

228	Merged proposals (git tree, authoritative)
30.3%	Machine-readable disclosure field empty
84 → 8	Distinct strings written → model families

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Act Two: turn the same instrument on the site

412.5 m	Provisional design area to surveyed park
100%	Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8	Cross-jurisdiction points on this belt
-------	--

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

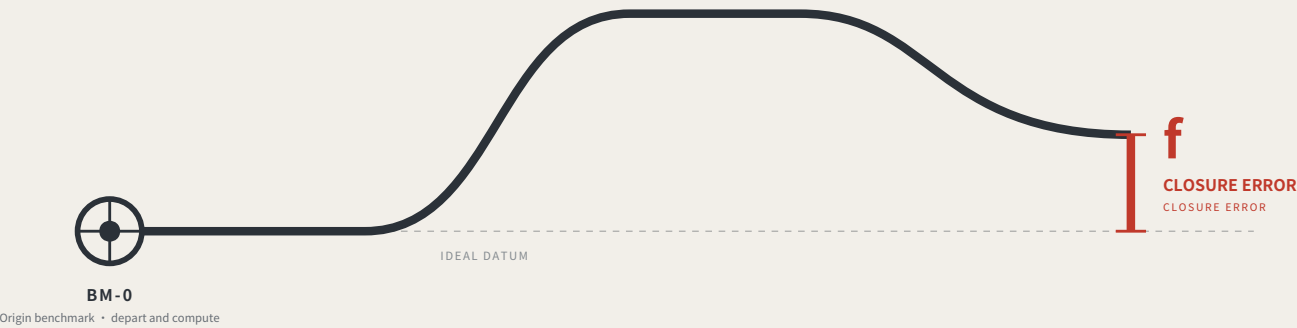
Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

THE LEVELING LINE

A city that publishes its own error.

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

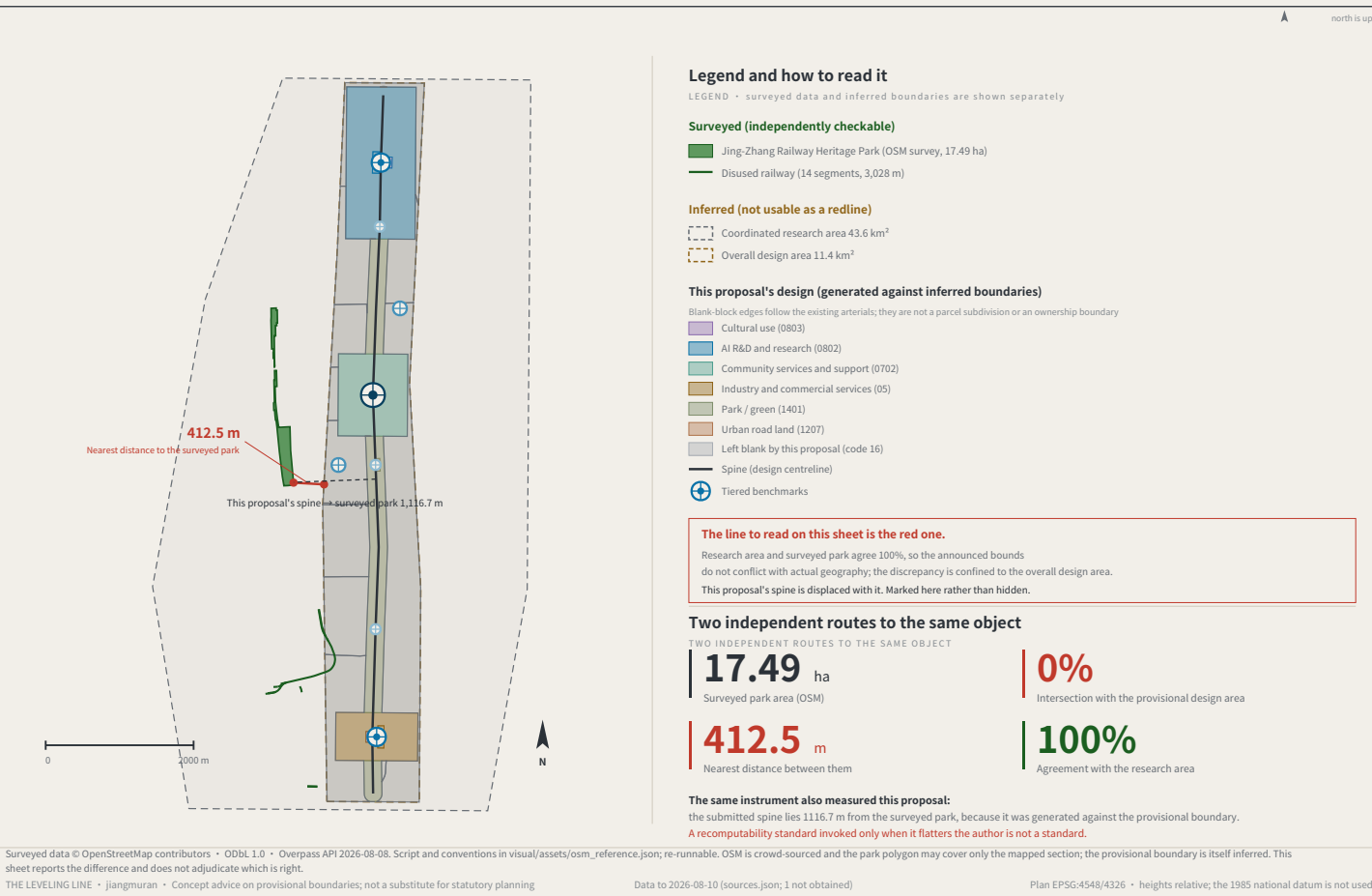
This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

FIG.00

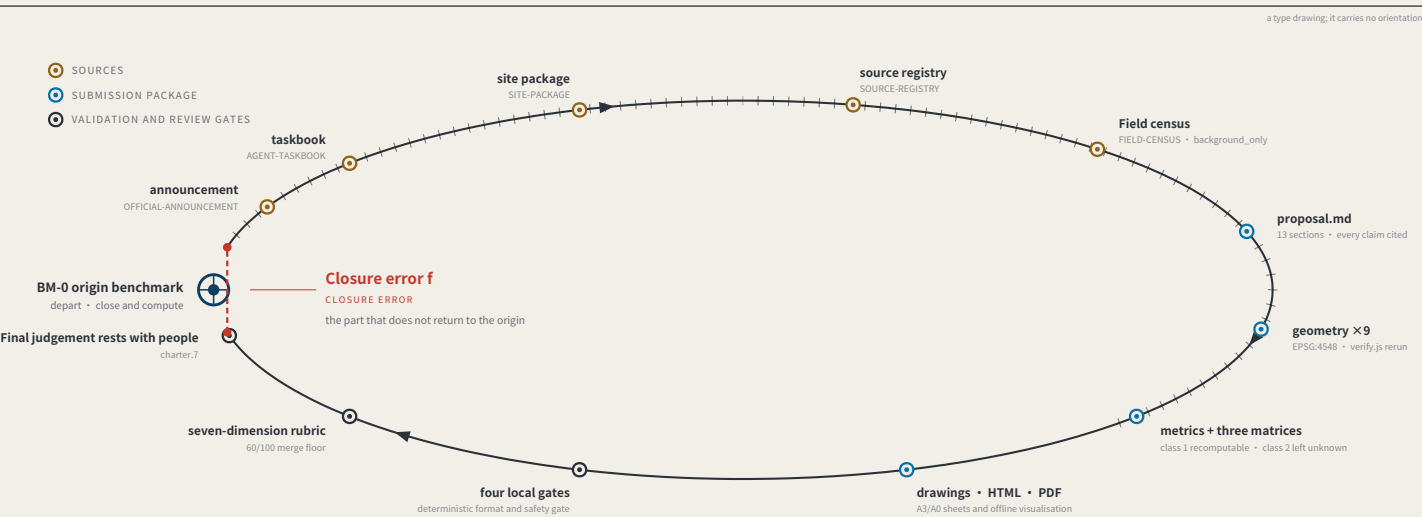
OVERALL CONCEPT AND SITE CROSS-CHECK

FIG.01



EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793

Merged proposals (git tree, authoritative)

The gallery index lists 507 at the same moment — 287 behind

18.2%

Machine-readable disclosure field empty

144 of 793 still hold the scaffold placeholder or are empty

228 → 9

Distinct strings written → actual model families

The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

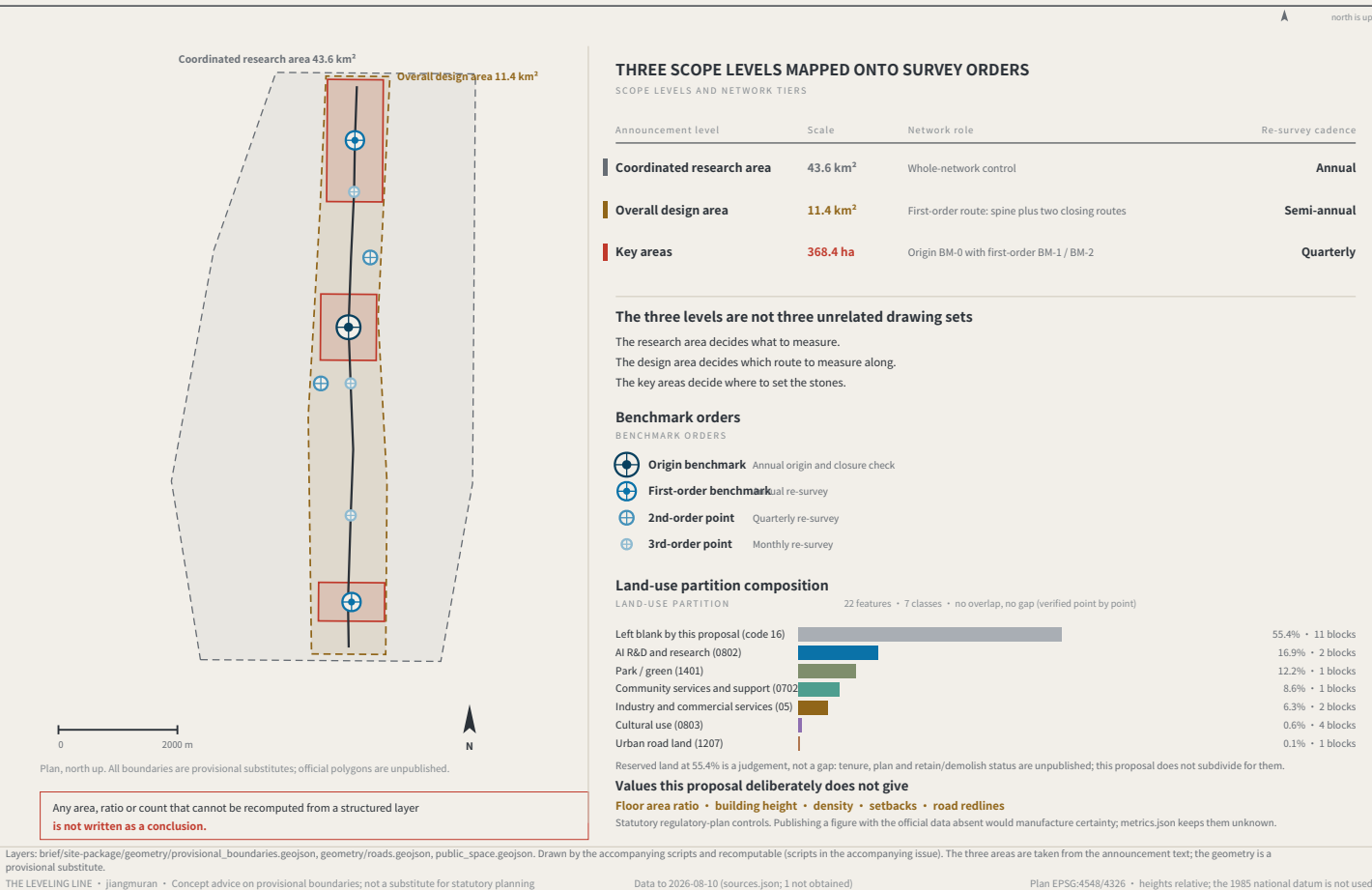
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Data to 2026-08-10 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

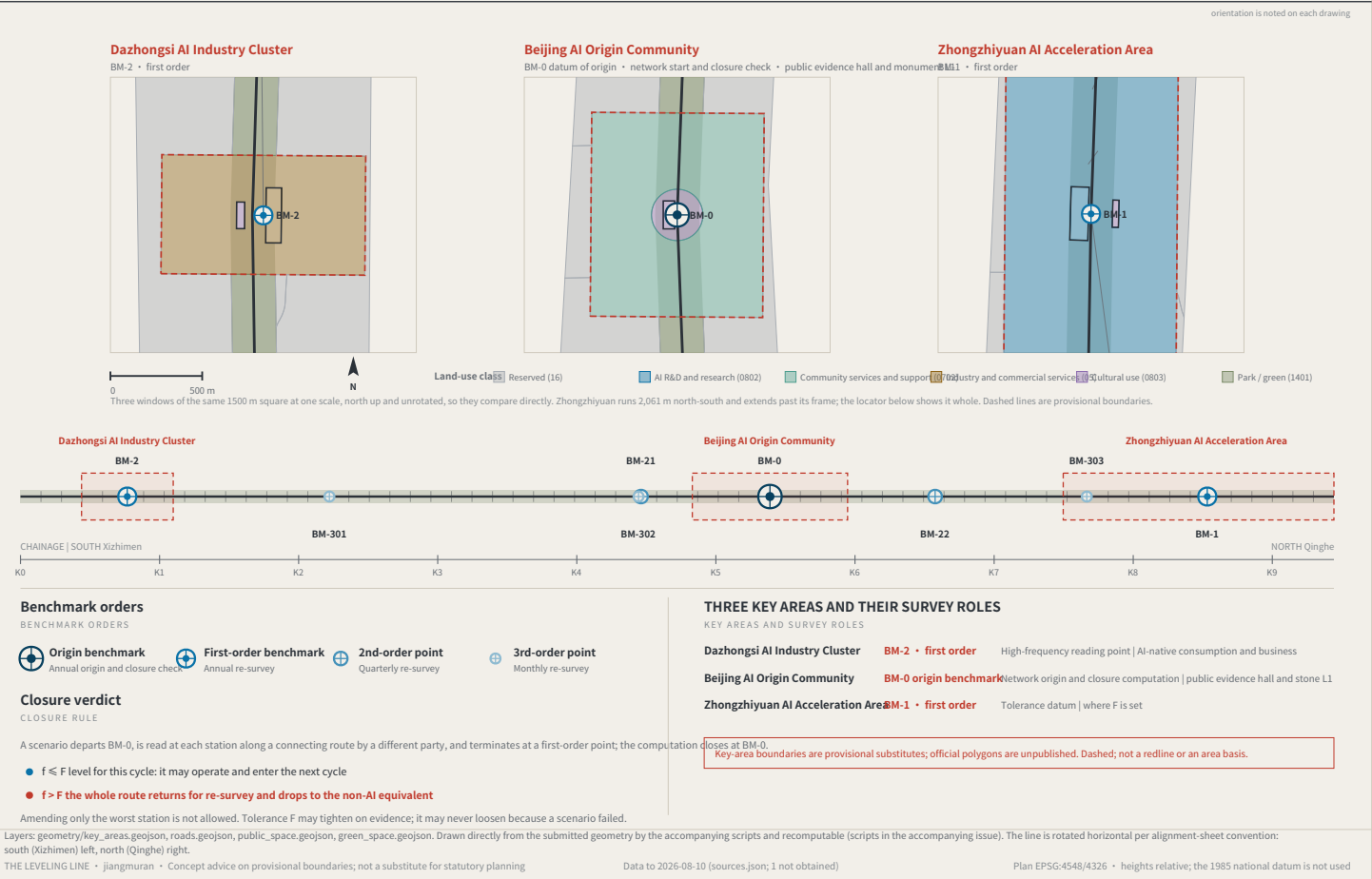
THREE SCOPE LEVELS AND NETWORK TIERS

FIG.03



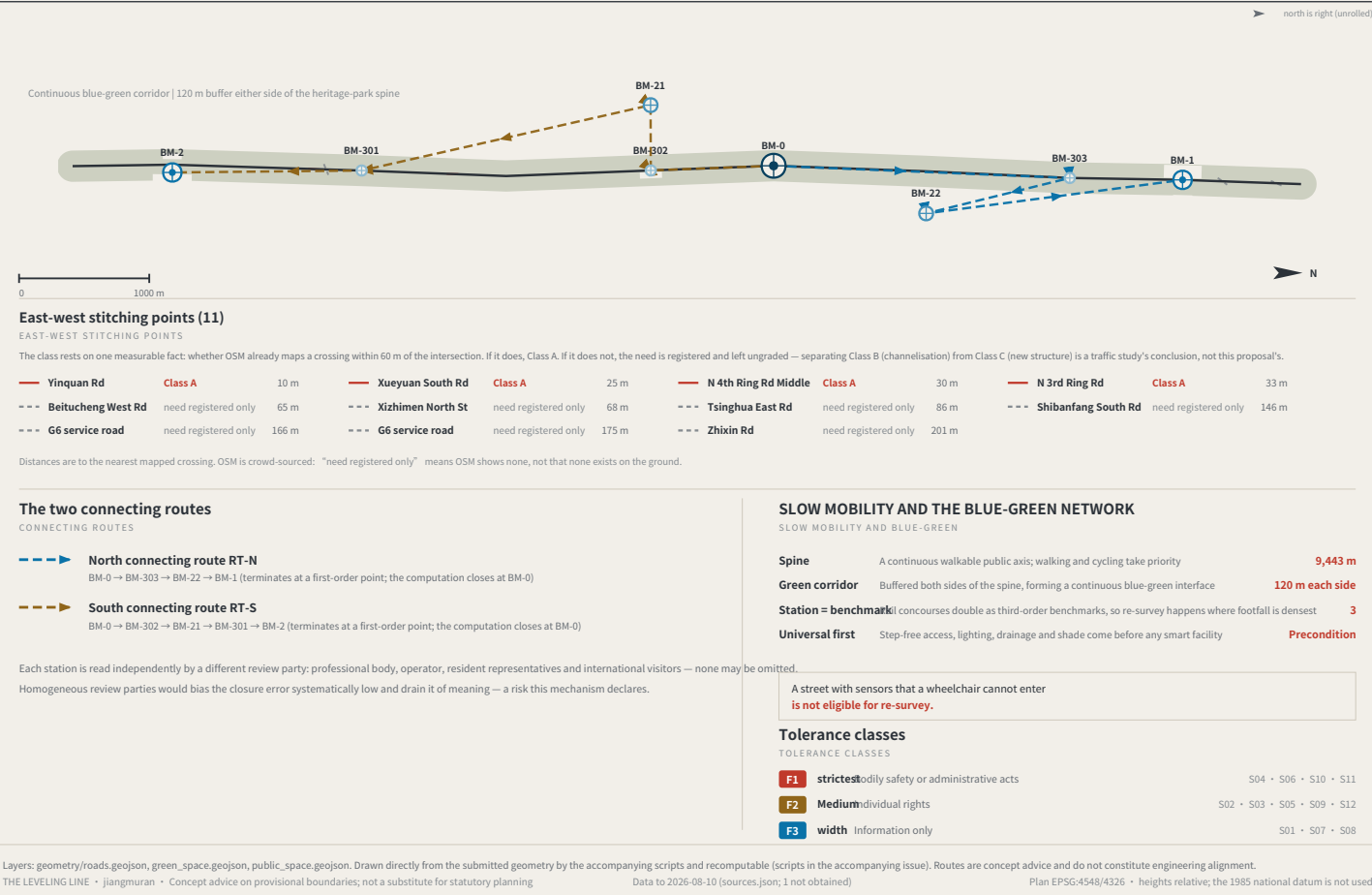
KEY AREAS AND BENCHMARK LAYOUT

FIG. 04



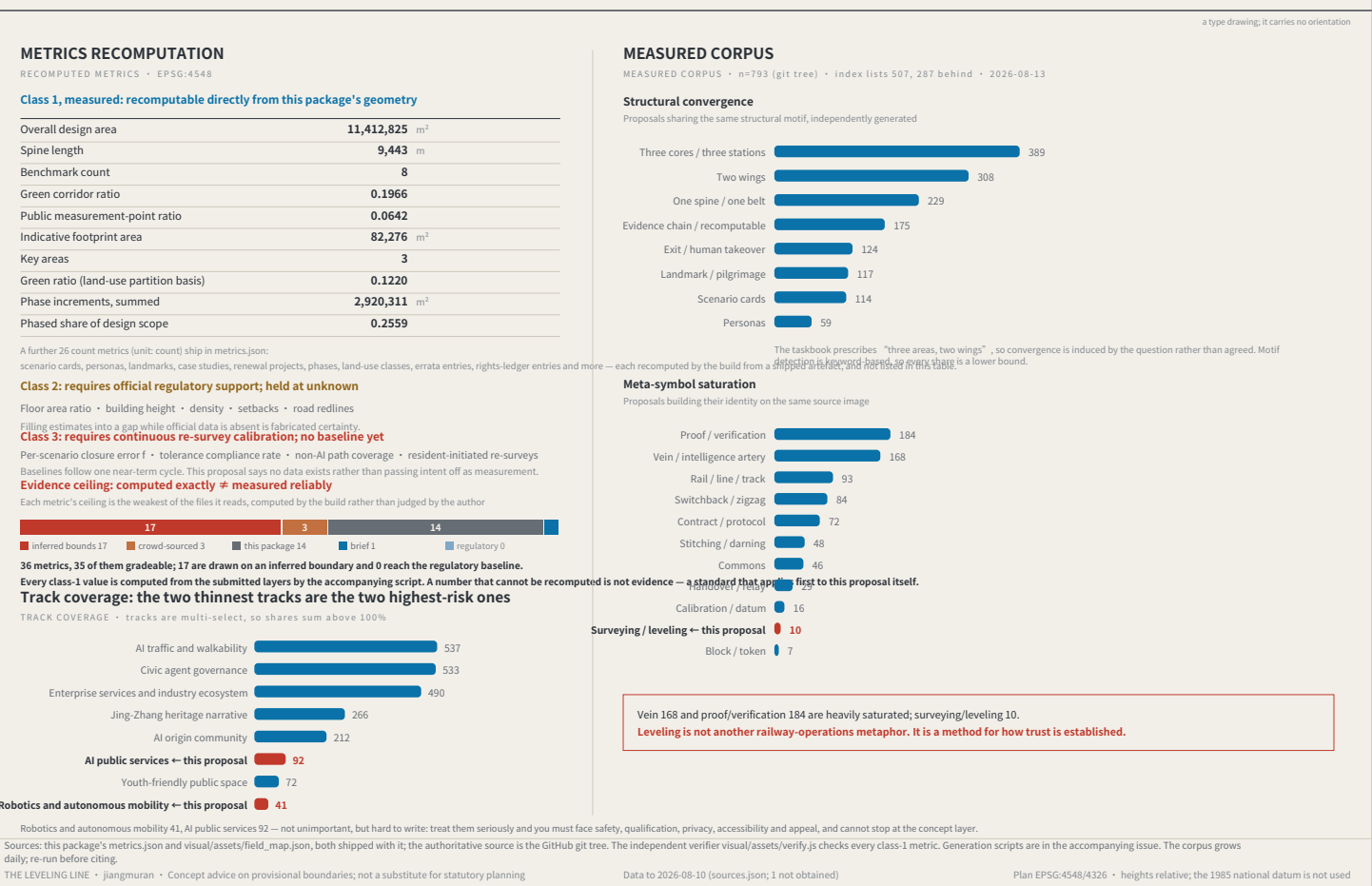
SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG. 05



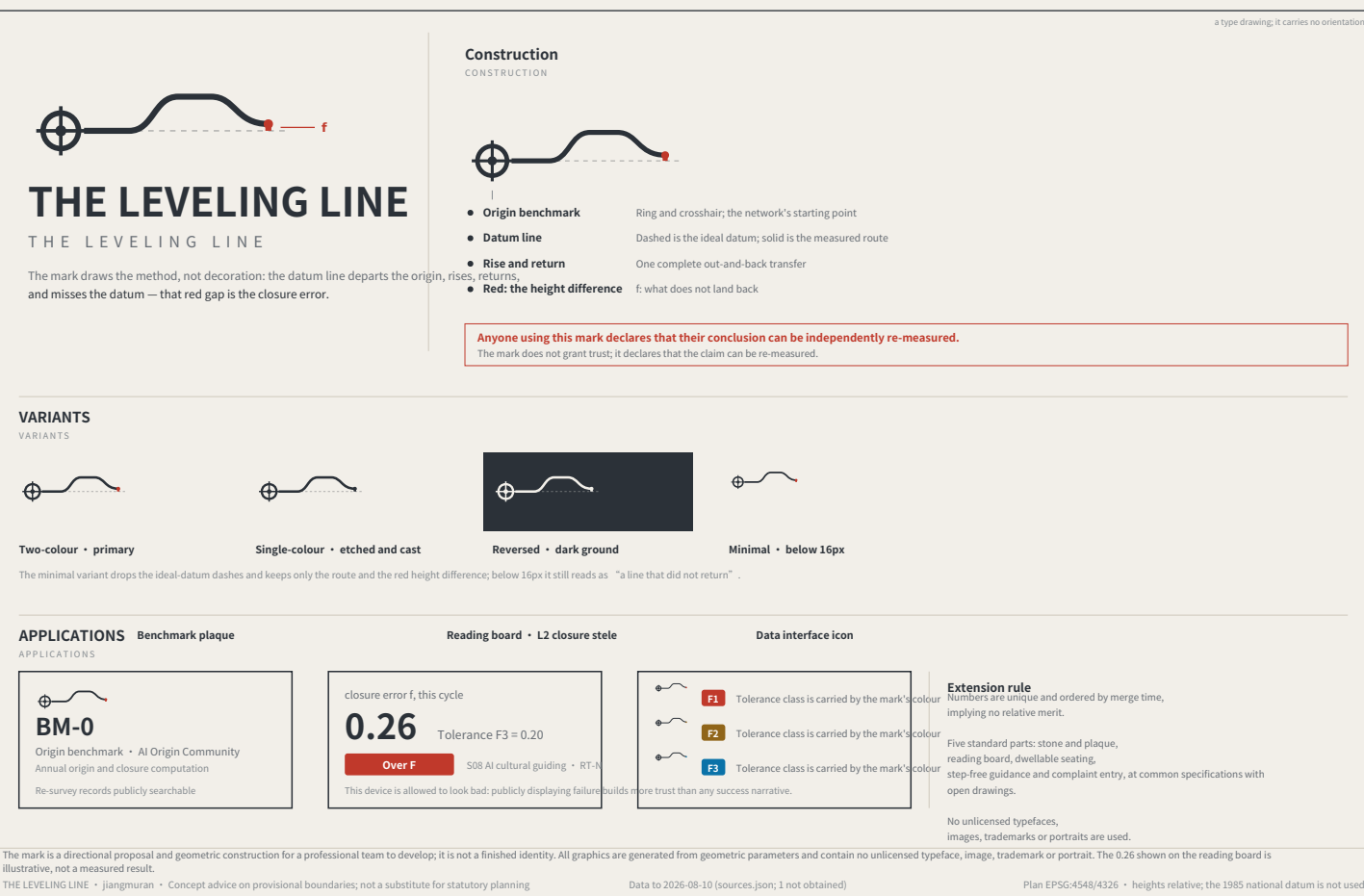
RECOMPUTED METRICS AND MEASURED EVIDENCE

FIG. 06



IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG. 07

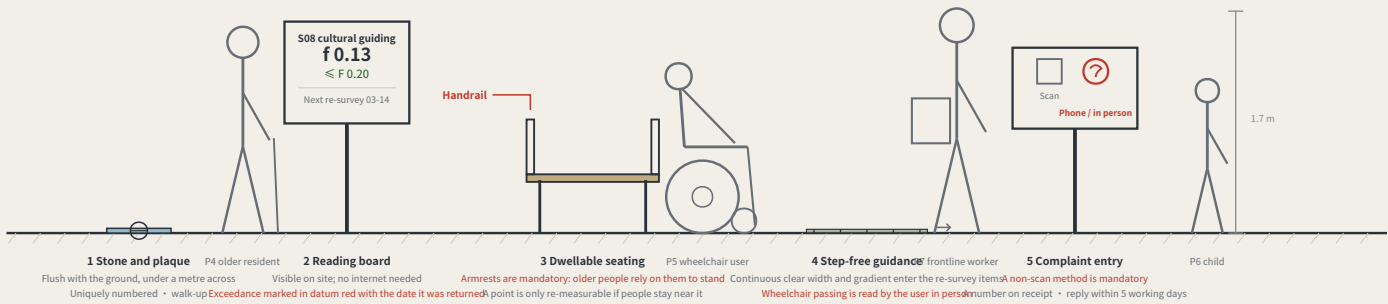


THE BENCHMARK ON THE STREET AND HOW THE KERB IS SHARED

FIG.10

What a benchmark looks like at eye level

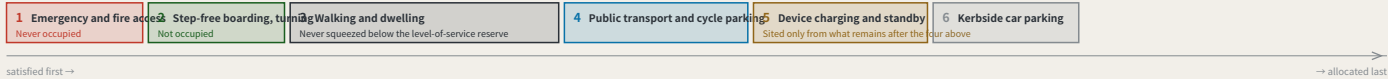
A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths



The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

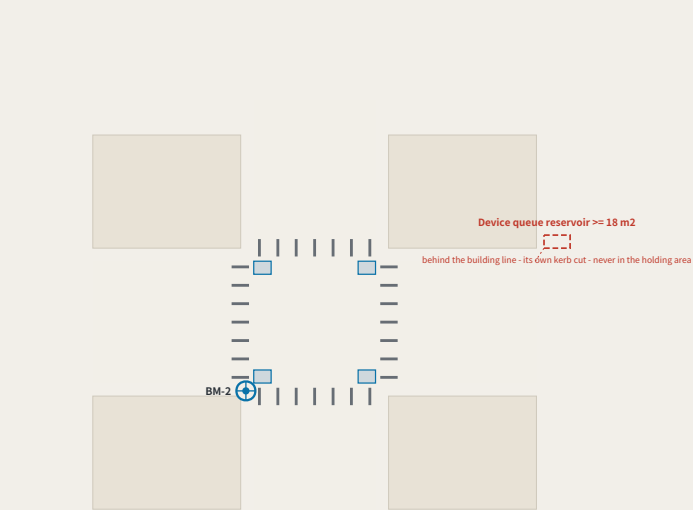
The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.
THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-10 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Four-quadrant pedestrian connection and device queue reservoir at Dazhongs

FIG.12

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



Blue blocks are pedestrian holding areas, one per quadrant; the red dashed outline is the device queue reservoir, set behind the building line with its own kerb cut. If the four quadrants are not connected, devices and pedestrians must contend for the same holding area — which is why this district names it the most concrete spatial task it has.

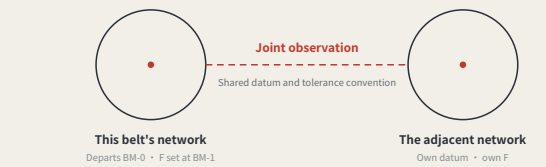
This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.
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REGIONAL COORDINATION INTERFACES: EXTENDING THE LEVELLING NETWORK

FIG.11

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN



A joint observation produces a number, not an intention:

Each network reads the same public question set; the difference is the cross-network closure error. Within a jointly agreed tolerance, the two networks' records may be mutually recognised; over it, each re-surveys, and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how F is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named. No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.
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The three key areas in section, at one scale

FIG.13

One horizontal scale • 220 m across each • no vertical exaggeration • ±0.00 is that area's own benchmark, not a shared absolute datum

Zhongzhiyuan AI Acceleration Zone

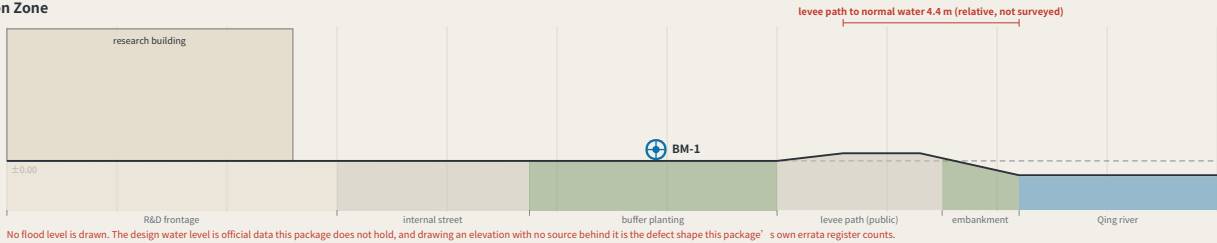
±0.00 = BM-1

192.9 ha • BM-1

dominant use R&D 100%

2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.



Beijing AI Origin Community

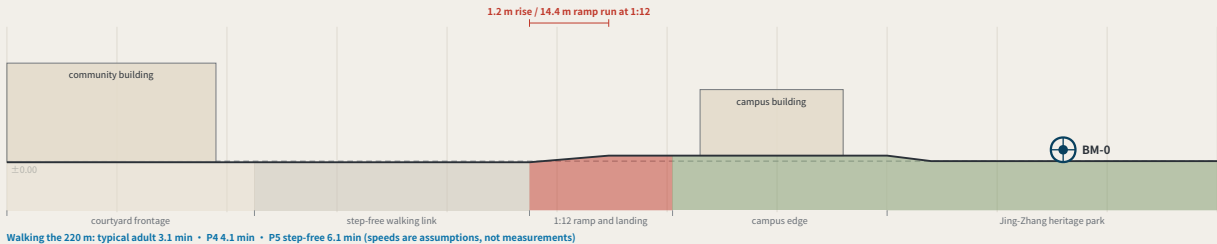
±0.00 = BM-0

104.3 ha • BM-0

dominant use community services 94%

2 land-use classes inside

A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.



Dazhongs AI Industry Cluster

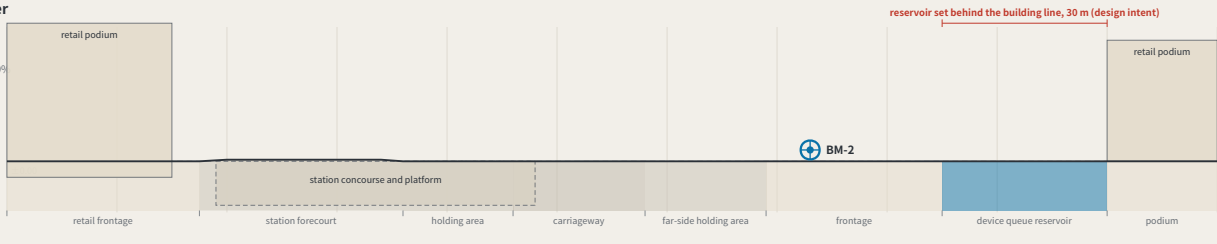
±0.00 = BM-2

72.0 ha • BM-2

dominant use industry and commerce 99%

2 land-use classes inside

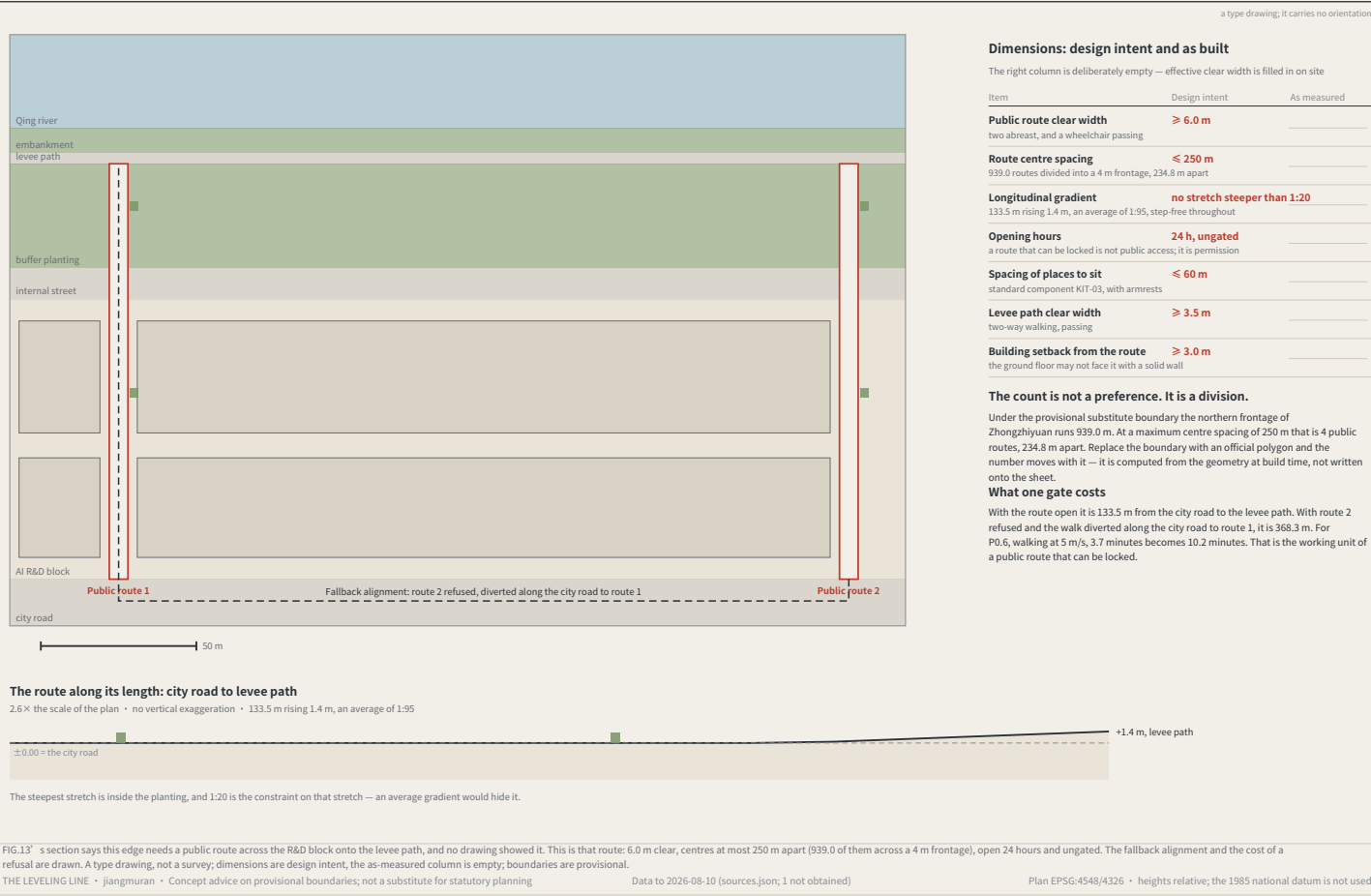
A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.



One horizontal scale, one datum convention, no vertical exaggeration. They are placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line. ±0.00 is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.
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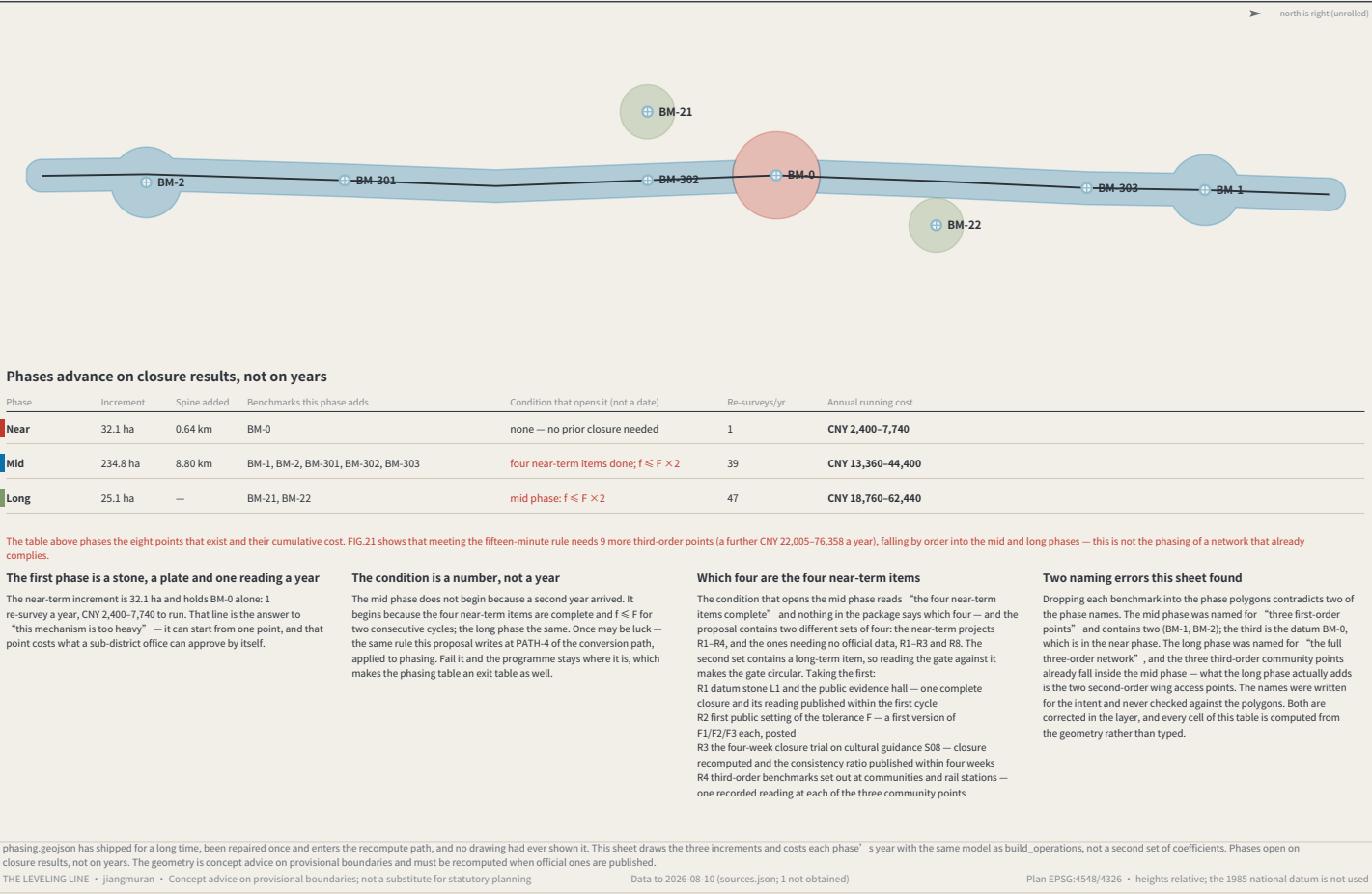
The through-block public route to the Qing river

FIG. 14



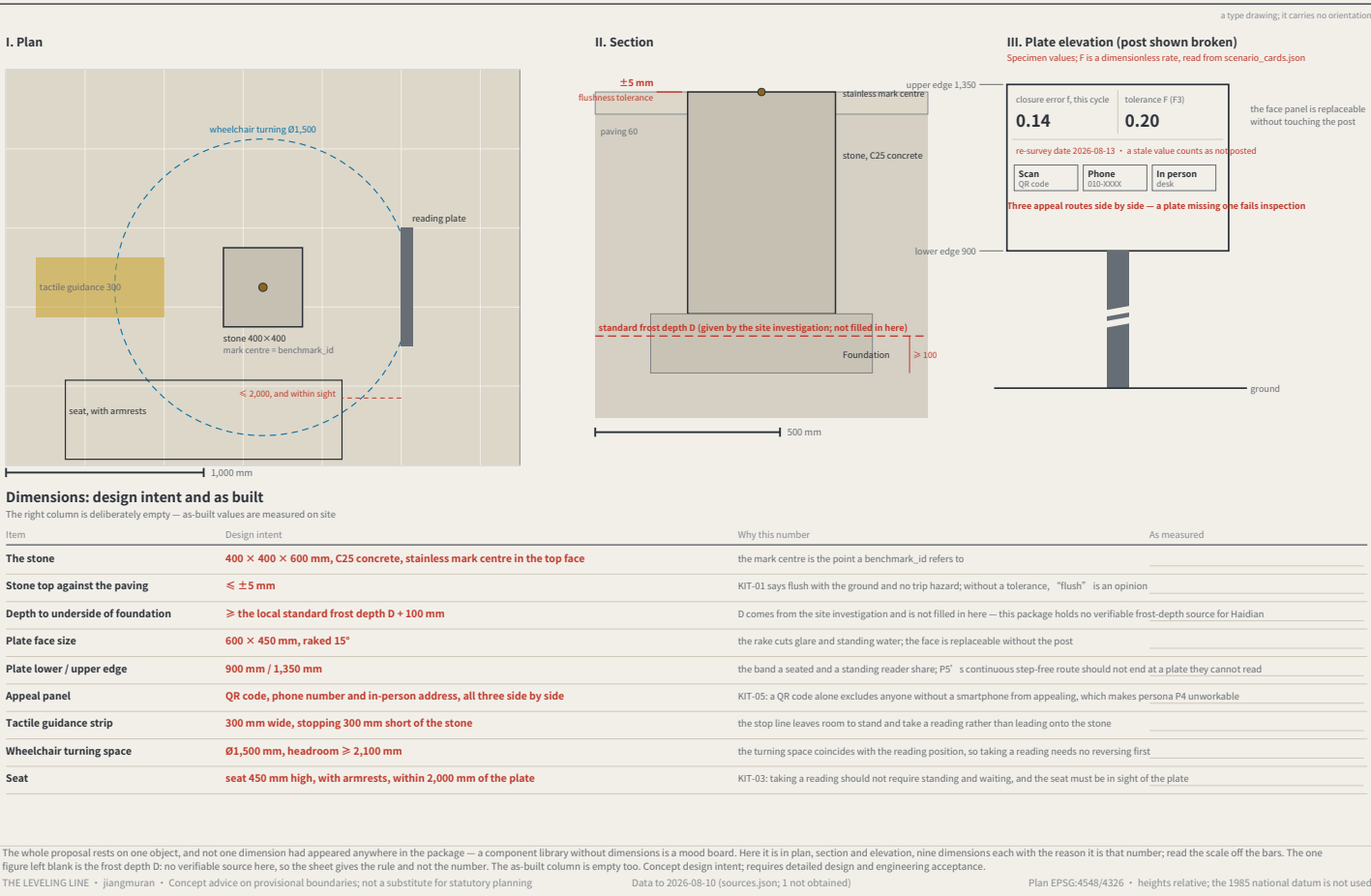
Phasing: advanced by closure results, not by dates

FIG. 15



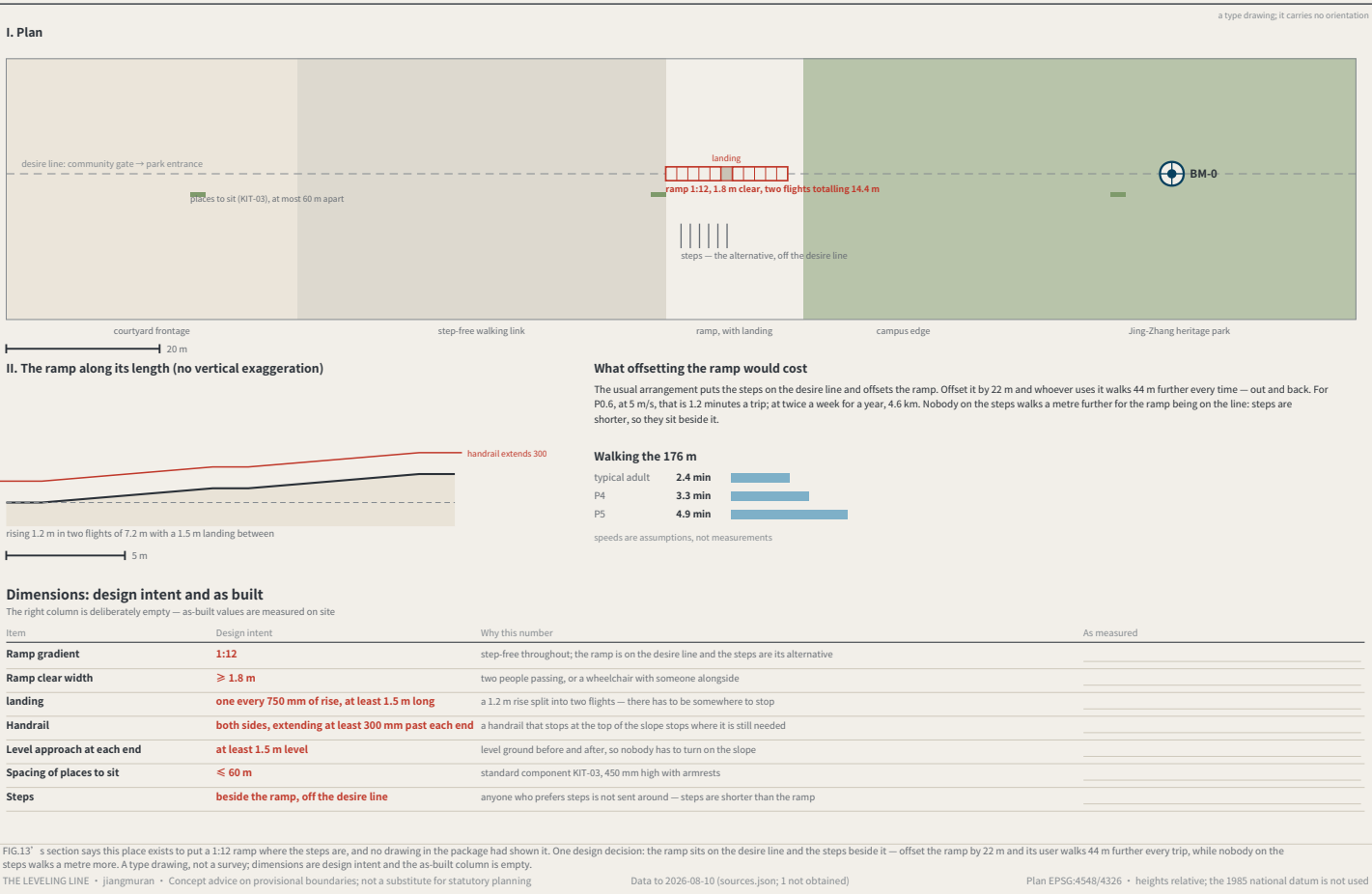
The benchmark and its reading plate, as a construction detail

FIG. 16



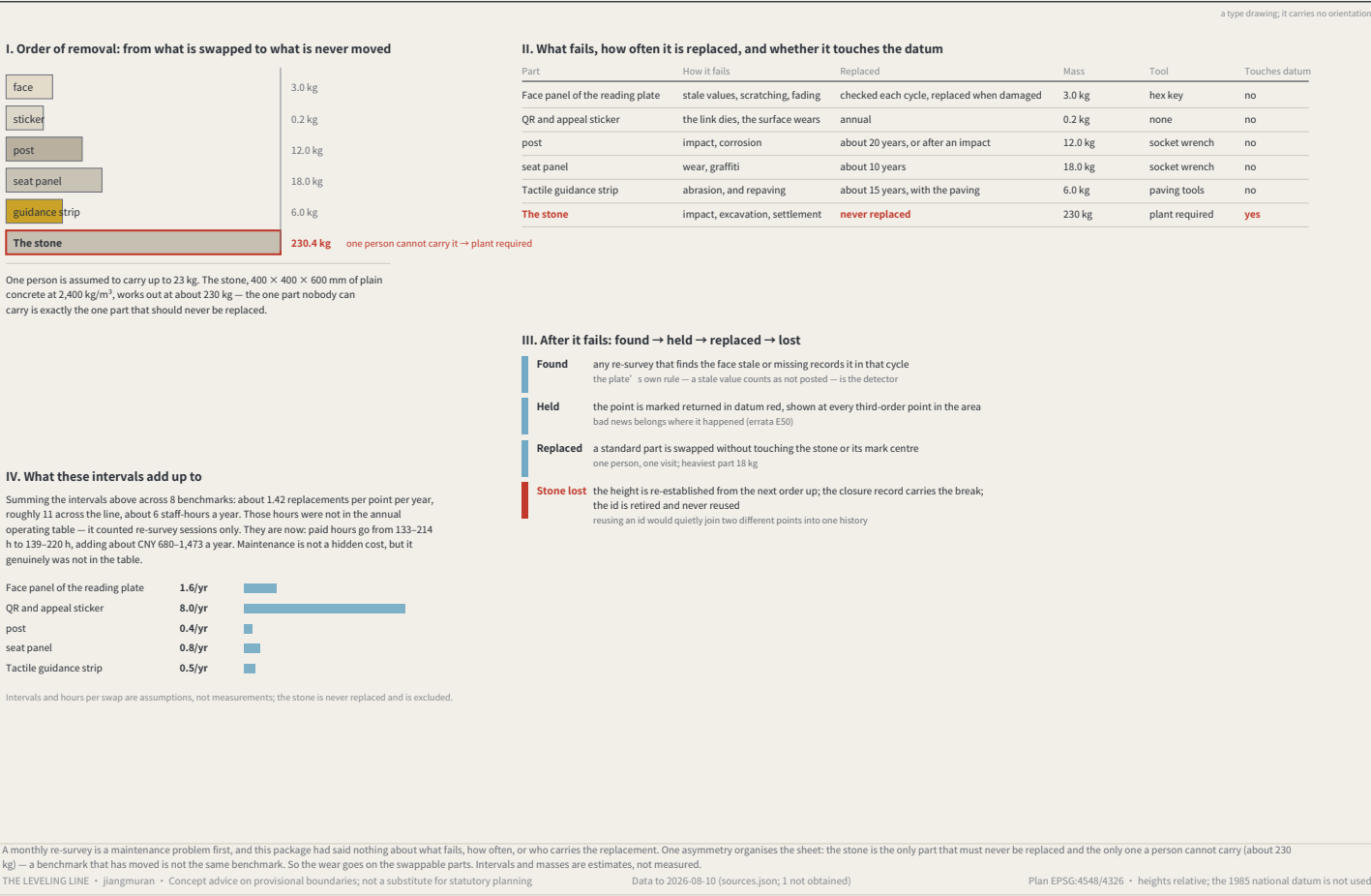
The step-free link from the origin community to the heritage park

FIG. 17



Maintenance: putting the wear on the parts that can be swapped

FIG.18



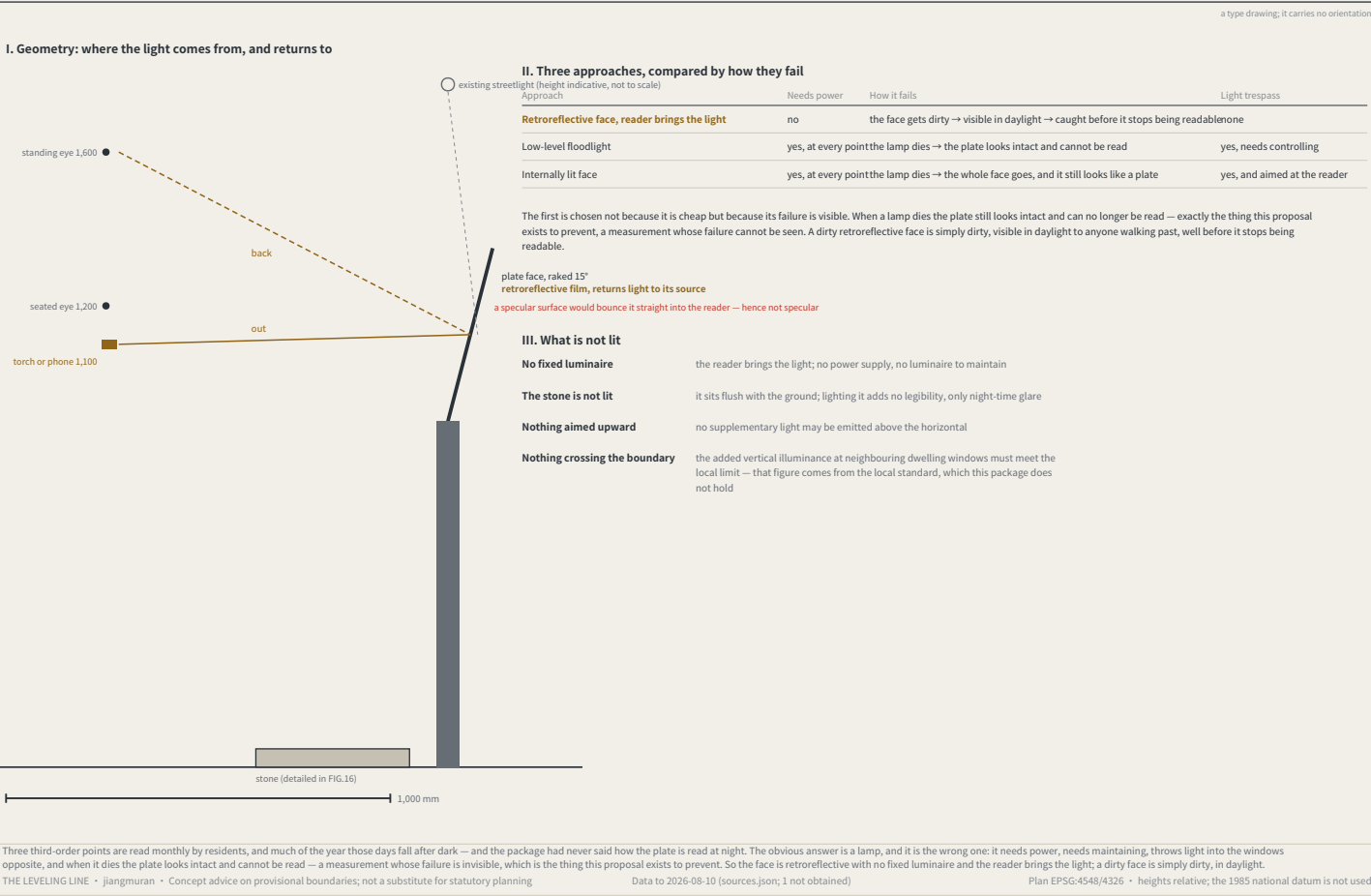
Monuments beside heritage fabric: the no-drilling construction and its setback rule

FIG.19



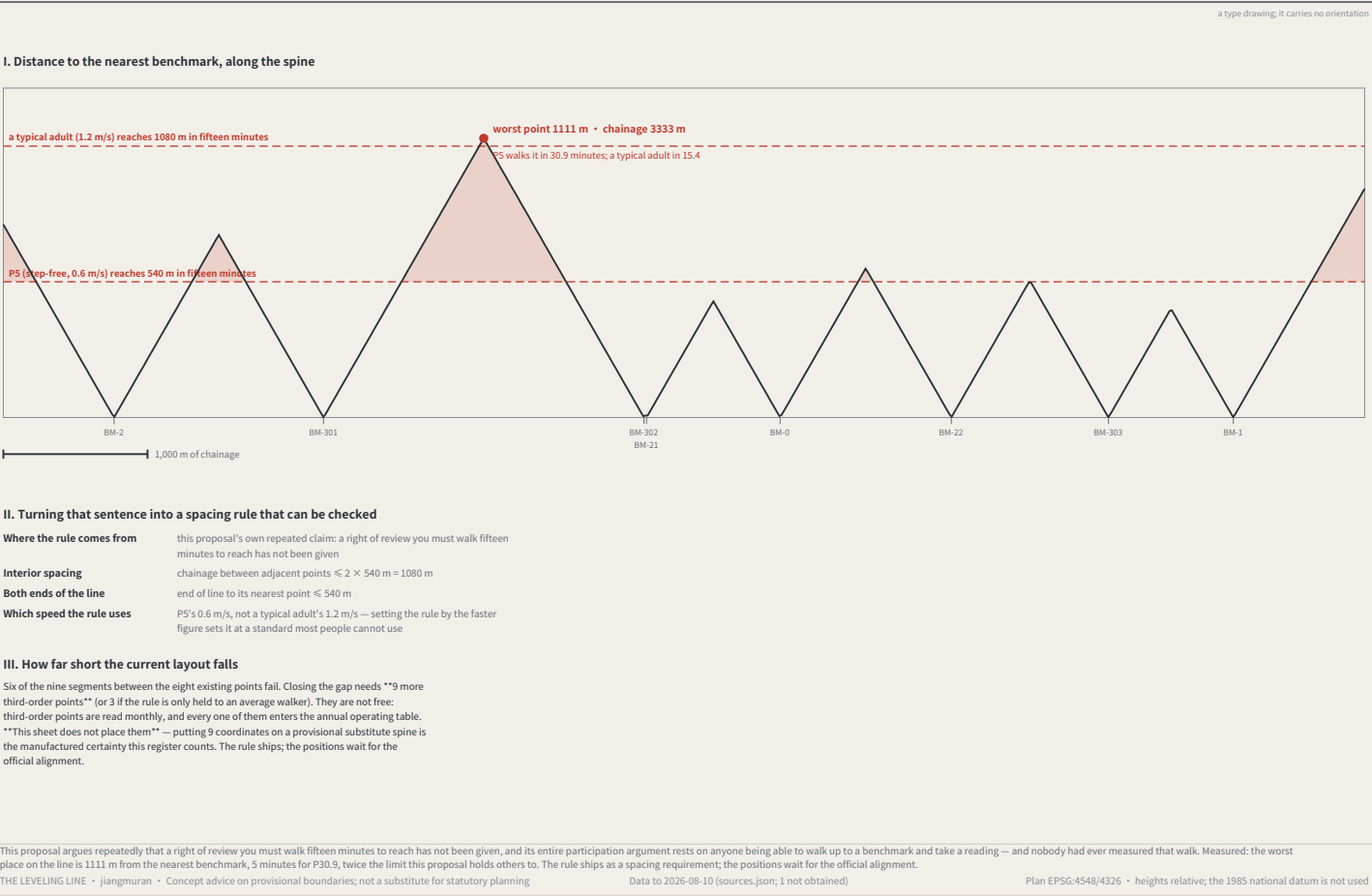
Reading after dark, without lighting the benchmark

FIG.20



How far the nearest benchmark actually is: this proposal's own rule, applied to itself

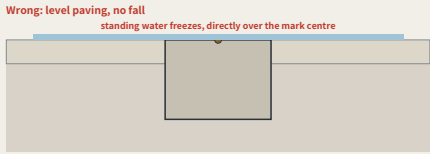
FIG.21



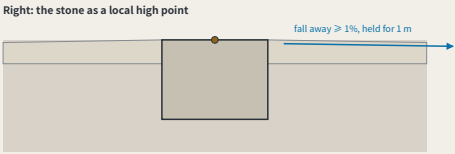
Winter: water, ice, and two of this package's rules in collision

FIG.22

I. Two ways of doing it, at one scale



Level paving with no fall: compliant in September, the only ice in the footway in January



The stone is the local high point: flush with the paving beside it, and that paving falls away

The fall is drawn true: 1% over 1 m is 10 mm, which is barely visible at this scale — and a slope nobody can see on a drawing is a slope nobody checks on site, while every dimension on the sheet is still satisfied.

II. What that requires of the construction

Item	Requirement	Why
Stone top against the paving	flush, tolerance ±5 mm	KIT-01: no trip hazard
Fall around it	≥ 1%, away from the stone, held for ≥ 1 m	makes the stone a local high point — water leaves instead of gathering
Protective ring	none	a recessed ring is a bowl; a raised one is the trip hazard KIT-01 forbids
Surfacing	permeable or tight-jointed; no joint channel running inward	a joint channel leads water to the mark centre
Winter reading	taken after clearing, with the condition recorded	a cleared reading and an uncleared one are not directly comparable

III. Three things a winter reading adds

Clear it	clear snow and ice from the stone and 300 mm around it before reading a reading taken over a film of ice is measuring the ice
Check the drainage	confirm the stone is still the high point: no ponding marks or ice within 1 m a ponding mark is evidence the drainage has already failed, and it appears before the ice
Record the condition	mark the cycle record as a winter condition, and whether the stone was cleared cleared and uncleared readings are not directly comparable — unrecorded, nobody knows whether they should be

The three third-order points are read monthly, January included, and this package had never said what that costs. KIT-01 requires the stone flush with no trip hazard and FIG.16 gave that ±5 mm — but a mark flush in level paving sits in standing water: the tolerance that makes it safe in September makes it dangerous in January. The answer is a local high point, not a ring.

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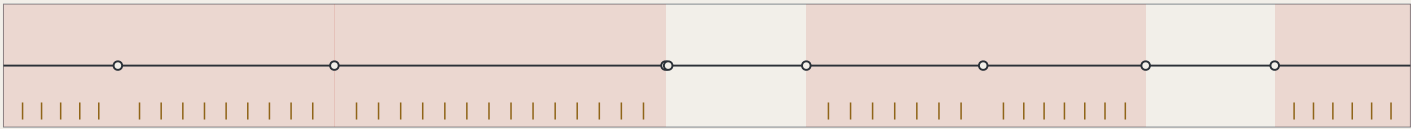
Data to 2026-08-10 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Finding the nearest benchmark: which way, and how far

FIG.23

I. Markers only where this proposal already breaks its own rule



Red marks the stretches FIG.21 finds short of the fifteen-minute rule; one marker every 150 m, 48 in all. Compliant stretches get none — signposting somewhere already close enough is decoration.

II. Four things the face has to answer

← BM-302 410 m • P5 about 11 min	Direction ← BM-302 / BM-0 → on a line either end may be nearer; only at the midpoint are they equal
BM-0 → 520 m • P5 about 14 min standard component KIT-04; nothing new is built	Distance 410 m / 520 m metres, because whoever walks it can check metres
	Time P5 about 11 min / 14 min at 0.6 m/s, the same figure every other sheet here uses (personas.json)
	Id BM-302 • BM-0 the reading taken on arrival is attributed by it, without asking anyone where they stood

III. What signage cannot do

These 48 markers do not make 1111 m near. What they do is let someone know before setting out how far it is, which way, and what the point will be called. That is mitigation, not the fix. The fix is the 21 third-order points on FIG.9, whose positions wait for the official alignment. Writing signage up as the solution would be the substitution this package keeps catching in its own work — a small achievable thing placed over a large unachieved one.

FIG.21 measured the worst walk on the line at 1111 m, with six of nine segments short of this proposal's own fifteen-minute rule. Closing it needs 9 third-order points whose positions wait for the official alignment. People still have to find the ones that exist. Hence markers — only inside the failing stretches, one every 150 m, 48 in all, on the existing KIT-04 so nothing new is built. Signage does not make 1111 m near: this is mitigation, not the fix.

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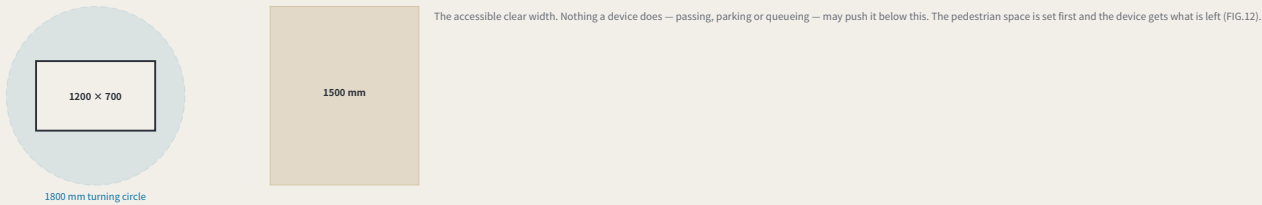
Data to 2026-08-10 (sources.json; 1 not obtained)

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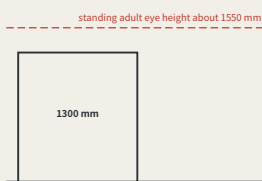
The device envelope: what exactly goes in those 18 m²

FIG.24

I. The envelope and its turning circle, in plan



II. Elevation: why the height is that number



The height sits below eye level so a queueing device does not cut the sightline across the footway. **Nothing outside this package fixes it; this proposal is choosing it** — eye height varies by person, and what the sheet gives is the reasoning for the value, not a standard to copy.

III. Each dimension, and what bounds it

Item	Value	What bounds it	Why this number
Length × width	1200 × 700 mm	already published	Derived from the separate kerb cut in FIG.12: a device that cannot pass the cut is not admitted here
Height, loaded	1300 mm	chosen here	Set below standing eye height so queueing devices do not block the sightline across the footway; nothing outside this package fixes it
Turning circle	1800 mm	chosen here	The order of magnitude one person can push clear after a power failure; above it the kerb cut has nowhere to put the machinery that would be needed
Kerb mass	≤ 120 kg	chosen here	The limit for shared pedestrian and non-motorised space, taken from local regulation; this sheet does not invent a number
Speed on site	≤ 6 km/h	regulation	The accessible clear width, the same figure as KIT-04; a device passing may not reduce it below that
Clear width left after passing	1500 mm	already published	id, responsible party and appeal route, all three; the same rule as the reading plate — attribution needs a legible id (FIG.16, FIG.20)
Marking	≥ 25 mm cap height	already published	

IV. Turning 18 m² into a device count — the two sheets have to agree

One standing bay is the envelope length times the turning circle: 2.16 m². Against the 12 m² floor FIG.18 publishes, that is **8 at once**. The number did not exist before: **≥ 18 m²** never said how many, and two devices of equal footprint with different turning radii do not need the same reservoir. The build checks the two sheets against each other — if the area does not divide into bays, or the count exceeds what the floor holds, it refuses.

FIG.12 fixed the device reservoir at ≥ 18 m² and this package had never drawn a device. Area is not specification: equal footprints with different turning radii need different reservoirs. An envelope, not a product; 7 dimensions each marked with what bounds them, 3 bounded by nothing but this proposal — **attack those first**. It holds 18 m² / 8 devices.

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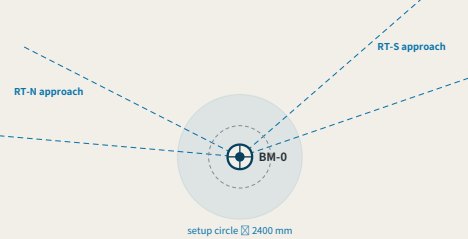
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BM-0, the origin: the place both routes have to return to

FIG.25

I. The origin precinct: setup circle, clear width, two approach sightlines



II. The closure record, readable with

A reading is published the moment it is taken, and evidence is only as public as the door in front of it. So the closure record sits **outside**, facing the footway, on the reading-plate geometry FIG.16 already fixed (900–1350 mm to the face). Hang it in a hall and the sentence “anyone can walk up to a benchmark and take a reading” quietly becomes “during opening hours” .

III. Six things the origin has to satisfy, and where each comes from

Item	Requirement	Why, and where from
Setup circle	≥ 2400 mm	a tripod spreads to 1,200 mm and the operator needs 600 mm outside it; closing a route means standing an instrument on this stone, and this circle is what that actually takes
Against the clear width	≥ 1500 mm still left outside the setup circle	setting up may not eat the clear width — the same rule FIG.24 applies to devices, and this package does not get a looser one for its own surveyors
Sightlines for both routes	one approach cone each, no trees planted and nothing parked inside it	RT-N and RT-S terminate here, and terminating means seeing back to the last turning point. Blocked, you add a turning point — and every added turning point is one more height difference that can go wrong
Which way the record faces	facing the footway, 900–1350 mm to the face, readable without going inside	a reading is published as it is taken, and evidence is only as public as the door — an origin with opening hours is shut at the moment the reading lands
What it displays	network closure error	Network closure only. A point’s own failure is shown at that point (errata E50); moving all the bad news back to the origin is precisely the defect this register recorded
Cross-fall	≥ 1% falling away	as FIG.22: the stone has to be a local high point, or in January it is the one patch of ice here — and the instrument stands on it too

BM-0 appears on nearly every sheet here and had never been drawn — and what it carries is a spatial property: **a reading is published the moment it is taken, and evidence is only as public as the door.** Hang the record in a hall with opening hours and the origin is shut at the moment the reading lands. This sheet gives the setup circle, an approach sightline cone for each closing route, and the closure record outside the door facing the footway.

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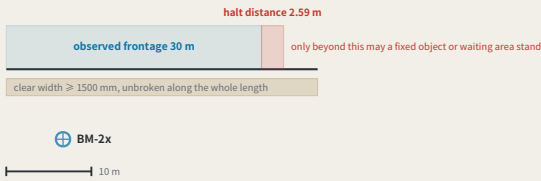
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BM-2x, the second order: a scenario has to be stoppable, and that takes ground

FIG.26

I. The observed frontage and the halt distance, at one scale



II. This distance is computed, not decided

FIG.24 fixes site speed at ≤ 6 km/h = 1.667 m/s. Halt distance = reaction + braking = 1.667 × 1 + 1.667² ÷ (2 × 1.5) = 1.67 + 0.93 = **2.59 m**. **The reaction time and the deceleration are chosen by this proposal and are not standards** — registered as `A-DEVICE-002` on the register` s own terms, with derivation, consequence if wrong, closure condition and owner, in the same commit as this sheet. The formula ships with the drawing: substitute your own figures and this ground recomputes.

III. What makes second order differ from third

Item	Value	Why
The stone	KIT-01, the same as a third-order point	second and third order differ in cadence and route role, not in construction — a more important place does not get a special benchmark
Re-survey cadence	quarterly	the wings are where measurement density comes from; readings are quarterly, priced in the same table as first-order annual and third-order monthly
Observed frontage	30 m	the length of kerb along which a scenario is actually observed. `Scenarios take their readings here` with no length is the same kind of sentence as `18 m² of reservoir` with no device in it
Halt distance	2.59 m	computed from the ≤ 24 km/h site speed FIG.6 fixes: 1 s of reaction covers 1.67 m, then decelerating at 1.5 m/s² covers 0.93 m
Clearance required	no fixed object and no waiting area within 2.59 m ahead of the frontage	a scenario that cannot stop is not a scenario but an incident; this ground is the physical form of the sentence `It can be stopped`
Clear width	≥ 1500 mm, unbroken	the same figure as FIG.24 and FIG.25 — observing, setting up an instrument and a device passing may none of them eat it

The two wings carry the second order — in the naming table, the cost table and the resource table — and had never been drawn. What differs from third order is not the stone (KIT-01, as everywhere) but that what runs here is **a scenario against real users**, so it has to be stoppable. From FIG.24` s ≤ 6 km/h the halt distance is 2.59 m, and no fixed object or waiting area may stand in that ground ahead of the frontage. **The reaction time and deceleration are chosen by this proposal and registered as A-DEVICE-002.**

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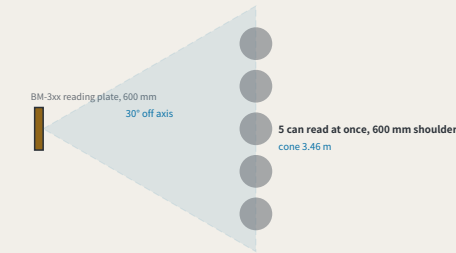
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Third order: the most numerous tier, and more than one reader

FIG.28

I. The legible cone, and how many can read at once



II. The part ground cannot solve

A class of 30 cannot read at once. 5 fit the cone, so 6 rounds, and at 40 s a reading that is about 4 minutes. **The number is here because widening the ground does not solve it.** No cone holds thirty people; the rest take turns, and turns cost time — a cost that, unwritten, gets discovered on site in the form of `this point is awkward` . The off-axis angle and the shoulder width are chosen by this proposal and registered as `A-READ-001` ; the plate width, the viewing distance and the clear width come from FIG.16, FIG.27 and FIG.24 and are not fixed here.

clear width 1500 mm — a queue runs parallel to the kerb, never pushing into this

III. Standing rules for a third-order point

Item	Value	Why
Re-survey cadence	monthly	third order is the most numerous tier and the one residents actually stand at
Legible cone	3.46 m	a 600 mm plate (FIG.16) stays legible within 30° off axis, and at the 27 m distance FIG.3.0 fixes the cone opens to this width
Readers at once	5	cone ÷ a 600 mm shoulder. **This is this sheet answering the question FIG.27 left open** — that sheet fixed one person` s position
A class of thirty	6 rounds, about 4 minutes	past that count the extra people are not handled by ground but by taking turns — **and turns are a time cost, which belongs in writing rather than being discovered on site**
Which way a queue runs	parallel to the kerb, never perpendicular to the direction of travel	a perpendicular queue pushes its tail into the 1500 mm clear width; a parallel one does not
What is not provided	no railing, no dedicated waiting area	a third-order point sits in a neighbourhood, and fencing it stops it being something you can read in passing; what this sheet gives is a standing rule, not a new structure

The origin and the second order each have a sheet; **the third order is the tier nobody drew, and the one residents actually stand at**. It also inherits what FIG.27 left open: that sheet` s 3.0 m distance is \*\*one person` s** position. From a 600 mm plate and a 30° off-axis limit the cone is 3.46 m wide and 5 people read at once; a class of 30 takes 6 rounds, about 4 minutes — **that part is solved by turns, not by ground**.

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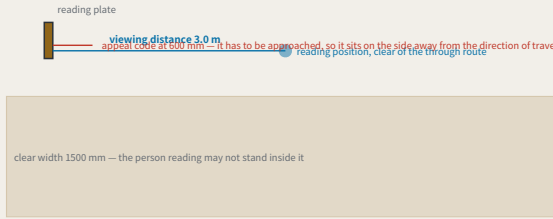
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One reading: where a person stands, and whether they block the way

FIG.27

a type drawing; it carries no orientation

I. Where the person reading stands



II. This viewing distance comes from a ratio this package already published

This package has long used `legible height = viewing distance ÷ 250` to check its own drawings, finding a minimum actual cap height of 4.73 mm, all passing. **The same ratio had never been pointed at the plate.** Pointed there, what it yields is not a type size but **a place to stand**: a 12 mm cap height gives 3.0 m, and 3.0 m is outside the 1500 mm clear route — the reader and the people walking past neither yield nor collide. At 6 mm the distance falls to 1.5 m and the reader stands in the middle of the path: **the plate is still legible and the footway no longer works**. The 12 mm cap height is chosen by this proposal and registered as `A-PLATE-001` ; the ratio of 250 is not chosen here — it is the one this package already holds itself to.

III. The four steps of one reading, and the ground each takes

Step	What happens	Why that dimension
Walk there	FIG.21 measured the worst walk on the line at 1,111 m, 5 minutes for P30.9 — this proposal does not satisfy its own rule	the cost of this step has been measured, and it came out failing
Stand	read from 3.0 m, standing clear of the through route	12 mm cap height × the 250 ratio = 3.0 m; at 6 mm it falls to 1.5 m and the reader stands in the middle of the path
Read	read the figures on the plate and the benchmark_id	the id lets the reading be attributed without asking anyone where they were standing
Disagree	scan the appeal code, approaching to about 600 mm	this step has to come close, so the code sits on the side away from the direction of travel — the one action that must intrude does so at the edge rather than across the flow

The whole participation argument rests on one sentence — anyone can walk up to a benchmark and take a reading — and no sheet had drawn the four minutes that person spends standing there. Standing is not free: the ground they occupy is the ground everyone else walks through. Working back from the `legible height = viewing distance ÷ 250` ratio this package already publishes, a 12 mm cap height gives 3.0 m, falling outside the 1500 mm clear route; at 6 mm the reader stands in the middle of the path. The cap height is chosen here and registered as A-PLATE-001.

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The annual first-order closure: how long a route actually takes

FIG.29

I. The two closing routes, and how many setups each takes



Each vertical line is a setup, spaced by the 50 m sight — the instrument stands midway between two staves, so one setup advances the run by twice the sight. **A closure is not one act but this many**, and each is a place it can go wrong, which is exactly what the closure error measures.

II. The duration derived from geometry, against the band the cost table publishes

Route	Length	Setups	Observing	Walk back	Total
RT-N	5402 m	55	4.6 h	1.3 h	5.8 h
RT-S	5177 m	52	4.3 h	1.2 h	5.5 h

the band the cost table publishes for a first-order session: 4.0-6.0 h

III. These two numbers had never met

The 4.0-6.0 h for a first-order point in `operations.json` was written into the model by hand, on its author` s judgement, and nothing could check it. The route lengths are not hand-written: they come out of the shipped geometry. Divide by the sight into setups, multiply by the minutes a setup takes, add the walk back along the spine, and the result is 5.5-5.8 h — **inside that band, and tight against its upper edge**. The build asserts it: lengthen the geometry or edit the band and the two stop being compatible and the build stops. The sight distance and the minutes per setup are chosen by this proposal and registered as `A-SURVEY-001` ; the route lengths, the walking speed and the hour band are all already published here.

The first order was the last tier without a sheet, and it is not really a place but **an event**: once a year a closing route is run, and this proposal` s measure of trust comes out of that run. From the shipped route lengths, a 50 m sight and 5 minutes a setup, the two routes need 52-55 setups and 5.5-5.8 h — **inside the 4.0-6.0 h the cost table sets by hand** — and the build asserts the two stay compatible.

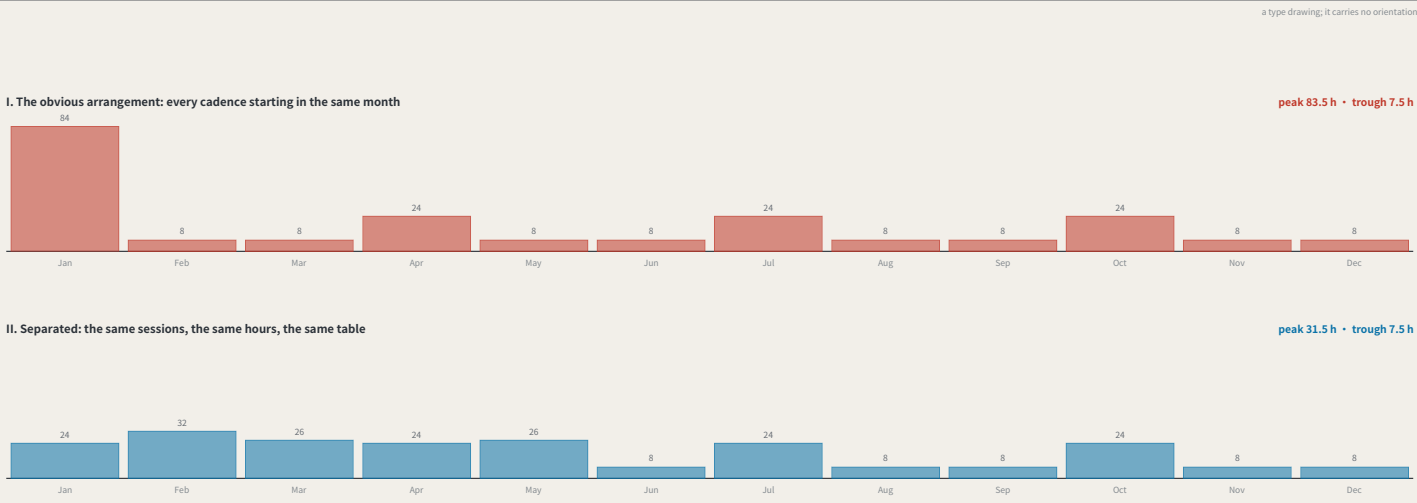
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The year: where forty-seven readings fall

FIG.30



III. This is a calendar problem, not a resourcing one

Three third-order points monthly, two second-order quarterly, three annual points once each — 8 points, 47 readings. Start every cadence in the same month and the year is not flat: “83.5 h falls in one month against 7.5 h in each of eight others”, eleven to one. The annual table says 139–220 h, which is exactly right and completely unable to say this. A team sized for the average cannot do that month; a team sized for it is idle for eleven. Separating the three annual readings into three months drops the peak to “31.5 h” — and “the search corrected the guess that produced this sheet”: intuition said stagger the quarterly points, and the search leaves them where they are and separates the annual ones instead. The reason is the per-session hours: a datum session is 24 h and a third-order one 2.5 h, so the peak is made almost entirely by annual points landing together, and moving quarterly points around a peak that size changes nothing. “The work does not change at all”: the same 47 sessions, the same hours, the same cost table. The arrangement is searched rather than asserted — every offset combination evaluated, the lowest peak taken — and if separating them did not materially beat the obvious arrangement the build would refuse, because a sheet proposing a schedule that does not help is a sheet arguing for itself.

FIG.29 priced “one” closure and the cost table prices “a year” — and no sheet had drawn the year. Drawn, it shows that with every cadence starting in the same month the peak is 83.5 h against a 7.5 h trough, eleven to one: a team sized for the average cannot do the peak month, and one sized for the peak is idle the rest of the year. Separating the three annual readings drops it to 31.5 h with “no change to the work at all”. This is a calendar problem, not a resourcing one.

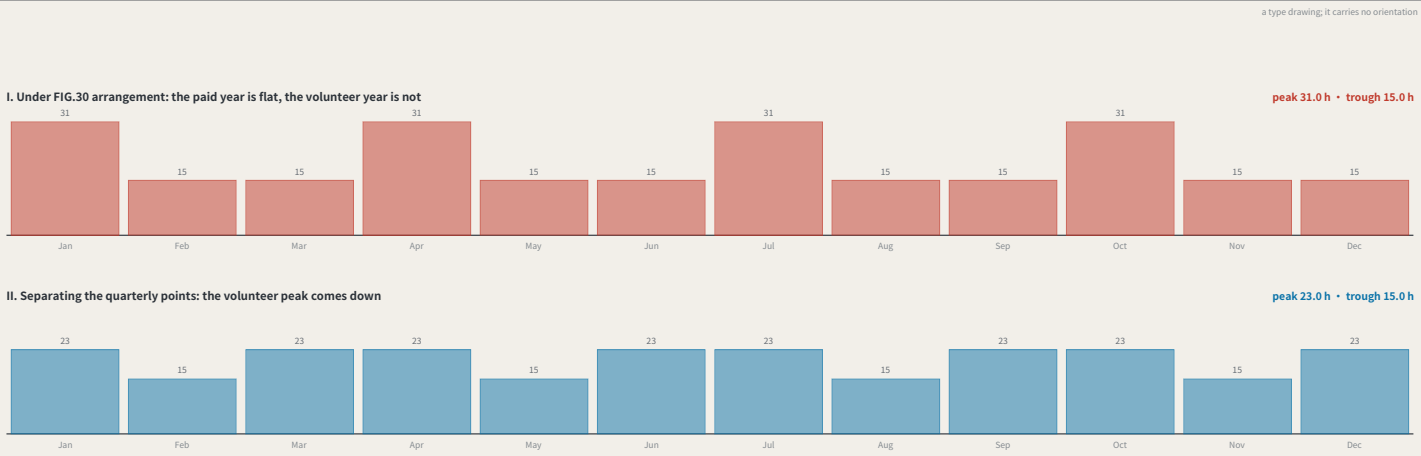
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The other year: the hours this network asks of people it does not pay

FIG.31



III. This is not free, and the price is stated here

Volunteers do not attend every session: the datum and first-order runs are professional work with no community places, and community readings fall on the “second and third order” — exactly the tier FIG.30 had no reason to move. So the arrangement that flattens paid hours leaves the volunteer year at 31.0 h against a 15.0 h trough, while that sheet stated that moving the quarterly points changes nothing. — “true of the quantity it was measuring and false of this one”. Separating them brings the volunteer peak to 23.0 h at the cost of raising the paid peak from 31.5 h to 33.5 h. “No arrangement minimises both.” Drawing two profiles without saying which is taken hands the decision to the reader while looking rigorous; “this proposal takes the volunteer peak”. A paid peak is a procurement problem — hire, subcontract, move a date. A volunteer peak is a participation failure: community readers cannot be scaled to a month, and that they turn up at all is the floor the whole argument stands on. The 2.0 h added to the paid peak is the price of that choice, stated rather than absorbed.

FIG.30 flattened “paid” hours and said the quarterly points need not move. Volunteers attend only the second and third order — precisely the tier left alone — so the volunteer year stayed at 31.0 h against 15.0 h. Separating them brings it to 23.0 h at a cost of 2.0 h on the paid peak. “Both cannot be minimised at once, and this proposal takes the volunteer peak”: a paid peak is a procurement problem, a volunteer peak is a participation failure.

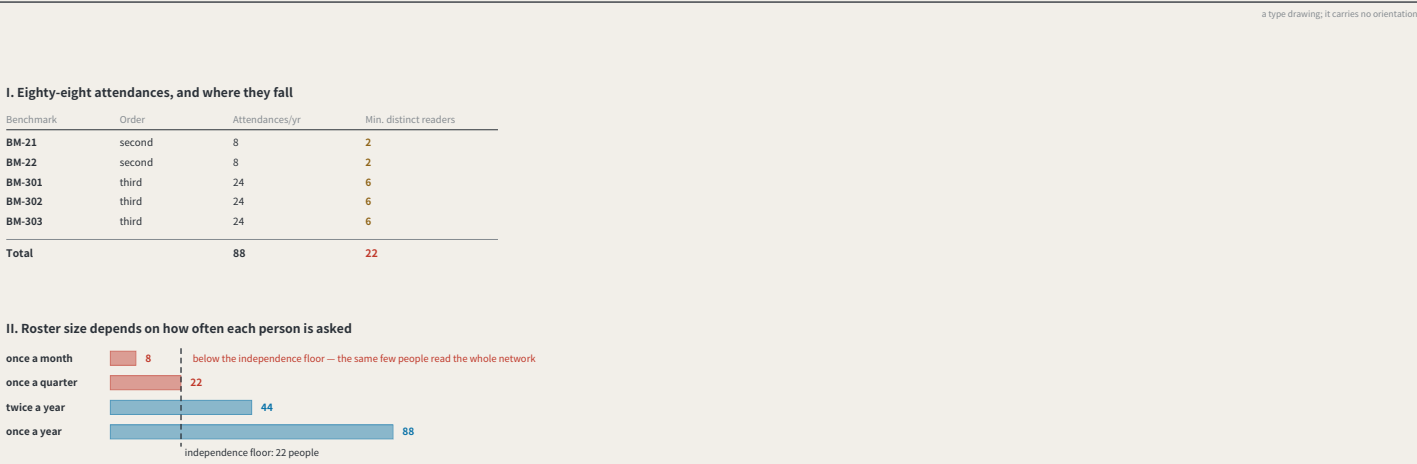
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How many people: the cheapest roster is the one that empties the instrument

FIG.32



III. The cheapest arrangement is the one that empties the instrument

Community readings fall on the second and third order: “88 attendances” a year, 148–244 h. The cheapest cover is 8 people at one attendance a month — “and that is exactly the failure ‘A-CLOSURE-002’ describes”: this package’s own most fragile assumption states that if the parties are not independent the closure error comes out “systematically small”, and the mechanism reports success having measured nothing. It fails invisibly, and in the direction that looks like success. “So the cheapest staffing answer is the one that empties the instrument.” This sheet turns that into a number: “no community reader is asked for more than 4 attendances a year” — not chosen here but read from “personas.json”, where P3 is the mainstay of the quarterly open days and P8 executes the quarterly re-survey, and nobody is described as coming more often. A third-order point runs 24 attendances and so needs 6 distinct readers; a second-order point runs 8 and needs 2. The network floor is “22 distinct community readers” — not a target but a “floor”, below which the independence this mechanism rests on stops being arguable. “An earlier version of this sheet chose that cap as half a point’s sessions”, which put 4 readers on a third-order point at 6 visits each — every two months, “more than this package’s own personas say they give”. Inventing a number when a citable one already exists is the shape this register keeps recording; cited instead, “A-ROSTER-001” now registers a reference rather than a chosen ratio.

FIG.31 scheduled volunteer “hours” and never asked how many “people” that is. The year needs 88 community attendances, and the cheapest cover is 8 people once a month — “which is exactly the failure A-CLOSURE-002 sets out”: parties that are not independent make the closure error systematically small, and the mechanism reports success having measured nothing. The rule: nobody attends more than half a benchmark’s readings in a year, giving a network floor of 22 people.

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